

The aim of the study was to determine frequency of hypoP, relationship between changes of serum phosphorus levels and clinical outcomes in adult critically ill patients with SARS-CoV-2.

Methods: The study included patients with a diagnosis of COVID-19 confirmed by RT-PCR, hospitalized in the intensive care unit with respiratory failure for more than 24 hours, and whose phosphorus value was recorded.

Results: A total of 205 patients were included. The median age was 64 (IQR: 54–75) years. The median APACHE II score was 17 (12–27). The most common comorbid diseases were hypertension (46%) and diabetes mellitus (29%). At admission and during 3-day follow-up, conventional oxygen support was provided to 33% of the patients, and 54 % of the patients were intubated and mechanically ventilated (MV). The frequency of hypoP in the first three days after admission to the intensive care unit was 25%. Patients baseline median serum phosphorus levels were 1.02 (0.87–1.25) mmol/L. Median ICU stay was 7 days and it was longer in HypoP patients ($p=0.046$). The cox regression analysis revealed that phosphorus levels in day 3 after ICU admission was identified as an independent predictor of ICU mortality (HR (95% CI): 1.46 (1.08–1.97)).

Conclusion: HypoP was seen in 25% in adult critically ill patients infected with SARS-CoV-2 in the first three days of ICU admission and, it was associated with increased ICU mortality and length of ICU stay.

Disclosure of Interest: None declared

P004-W

EFFECTS OF TRANSCUTANEOUS ELECTRICAL ACUPOINT STIMULATION (TEAS) ON POSTOPERATIVE RECOVERY IN PATIENTS WITH GASTRIC CANCER: A RANDOMIZED CONTROLLED TRIAL

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Rationale: Transcutaneous electrical acupoint stimulation (TEAS) is an innovative choice for postoperative pain management. However, the safety and effectiveness of this traditional Chinese medicine (TCM) therapy for patients who underwent gastrectomy is largely unknown. So, the purpose of this study is to evaluate the safety and effectiveness of TEAS for patients who underwent gastrectomy.

Methods: We recruited 96 patients with gastric cancer from May 2019 to November 2019; 82 patients were enrolled, and 81 patients completed. Patients were randomly assigned to TEAS group (TG) received TEAS on postoperative day (POD) 1–3 or control group (CG) at a 1:1 ratio. The primary outcomes were pain score and consumption of analgesics. The secondary were the time of first postoperative flatus and defecation, frequency of postoperative nausea, vomiting, distention, diarrhea, comfort of semi-fluid diet, Clavien-Dindo grade (C-D grade) and length of postoperative day. We performed hematological analysis to explore the possible mechanisms.

Results: Overall, 81 patients were enrolled included in the analysis. Compared with CG, pain scores in TG were lower on POD 1–5, and the use rate of opioids was lower; time of first postoperative flatus and defecation were shorter; the frequency of nausea were fewer and patients were more comfortable with semi-fluid diet; among the hematologic results, β -endorphin (β -End), interleukin-2 (IL-2), motilin (MTL) on POD 3, POD 5 were lower, 5-hydroxytryptamine (5-HT), interleukin-6 (IL-6), tumor necrosis factor- α (TNF- α) were higher. And no adverse event was reported.

Conclusion: TEAS can relieve postoperative pain and promote the recovery of gastrointestinal function. Consequently, it can be an adjunctive therapy to enhance postoperative recovery for patients after gastrectomy.

Disclosure of Interest: None declared

P005-W

ENHANCED RECOVERY AFTER SURGERY IN ELDERLY GASTRIC CANCER PATIENTS UNDERGOING LAPAROSCOPIC TOTAL GASTRECTOMY

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Rationale: The aim of this study was to evaluate the effects of the enhanced recovery after surgery (ERAS) program versus conventional perioperative care on the short-term postoperative outcomes among elderly patients with gastric cancer who are undergoing laparoscopic total gastrectomy.

Methods: Elderly patients with gastric cancer who are undergoing laparoscopic total gastrectomy were randomized to ERAS or conventional perioperative care groups. Short-term postoperative outcomes, including postoperative hospital stay, mortality, complications, readmission rate, and reoperation rate were compared between the two groups. In addition, blood samples were taken preoperatively and on postoperative days 1, 3, and 5. Systemic human leukocyte antigen (HLA)-DR expression on monocytes and C-reactive protein (CRP) were analyzed.

Results: Of the 171 eligible patients, 85 patients were assigned to receive ERAS program treatment and 86 patients to receive conventional care. The patients' characteristics were comparable. Postoperative hospital stay was shorter in the ERAS group than in the conventional group. Hospital mortality, overall morbidity, morbidity Clavien-Dindo (C-D) grade II, readmission rate, and reoperation rate did not show significant differences between the two groups. However, morbidity C-D grade IIIa was lower in the ERAS group than that in the conventional group. The ERAS program shortened the number of days to postoperative first flatus, first defecation, semifluid diet, and soft bland diet. Moreover, the ERAS program increased the HLA-DR expression on monocytes and decreased the CRP levels on postoperative days 1, 3, and 5.

Conclusion: The ERAS program was feasible and effective for elderly patients with gastric cancer who are undergoing laparoscopic total gastrectomy. The benefits of ERAS were associated with improvement of impaired immune function and suppression of inflammatory reaction.

Disclosure of Interest: None declared

P006-W

THE EFFECT OF PERIOPERATIVE ERAS PATHWAY MANAGEMENT ON SHORT-AND LONG-TERM OUTCOMES OF GASTRIC CANCER PATIENTS

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Rationale: To compare postoperative short-term outcomes and long-term prognosis between perioperative Enhanced Recovery After Surgery (ERAS) and conventional pathways protocols in gastric cancer patients.

Methods: This is a single institute retrospective cohort study, all patients were pathologically proved to be gastric adenocarcinoma, underwent standard radical gastrectomy with D2 lymphadenectomy during the period of 2007–2012. Total 2124 cases were eligible to be analysed and divided into ERAS groups and Non-ERAS group according to the different perioperative pathway protocol. Propensity score matching method was used to balance the baseline characteristics. The short-term clinical outcomes and overall 5-year survival rates were compared between the two groups.

Results: The incidence of overall postoperative complications was similar between the two groups, including anastomotic leakage, abdominal hemorrhage, etc. But the incidence of SSI, atelectasis, and thromboembolic disease in ERAS group was significant lower than that in Non-ERAS group. The number of lymph node harvested, operation time, intraoperative blood loss, postoperative hospital and cost in ERAS group were better than those in non-ERAS group. There were no significant differences in unplanned reoperation, 30 day readmission rate of discharge and postoperative mortality between the two groups. The 5-year overall survival rates of non-ERAS group and ERAS group were 66.2% and 72.8% respectively. The subgroup analysis found that 5-year OS rates of stage I were